Caution. The relay contacts may carry mains voltage. Provide proper insulation, enclosure, earthing, fuse protection, and adequate PCB spacing. Never touch the circuit while it is connected to the mains.

Alternative modified off-delay timer

Fig. 7 shows the alternative off-delay timer, which is a modification of the off-delay timer shown in Fig. 4. This is also a 12V off-delay timer using the CMOS CD4049 inverter. When the 12V supply is switched on, C1 starts charging through R2 and VR1. The voltage at the input of IC3A gradually rises. Until this voltage reaches the switching threshold of the 4049, the inverter output remains high, keeping T1 conductive. The timing can be adjusted with VR1, while S1 provides a reset/discharge path for C1. IC3 (IC3A-IC3G) has six inverters, but only IC3A is used.

PARTS LIST (FIG. 4 OFF-DELAY TIMER)

Semiconductors:
IC2
(IC2A-IC2G) - 4049 hex inverter
T1 - 2N3904 NPN transistor
D1, D2 - 1N4007 rectifier diodes
LED1, LED2 - 5mm red and green LEDs
Resistors (all 1/4-watt, ±5% carbon):
R1, R4 - 4.7kΩ
R2, R3 - 1kΩ
VR1 - 1MΩ potmeter
Capacitors:
C1 - 470µF, 25V electrolytic
C2 - 220µF, 16V electrolytic
Miscellaneous:
RL1 - 12V SPDT relay
CON1, CON2 - 2-pin connectors
BATT.1 - 12V battery/12V SMPS supply
LOAD - Anything like geyser or water pump

PARTS LIST (FIG. 7 ALTERNATIVE OFF-DELAY TIMER)

Semiconductors:
IC3
(IC3A -IC3G) - 4049 hex inverter
T1 - 2N3904 NPN transistor
D1, D2 - 1N4007 rectifier diodes
LED1, LED2 - 5mm red and green LEDs
Resistors (all 1/4-watt, ±5% carbon):
R1, R5 - 1kΩ
R2, R4 - 4.7kΩ
R3 - 10kΩ
VR1 - 1MΩ potmeter
Capacitors:
C1 - 470µF, 25V electrolytic
C2 - 220µF, 16V electrolytic
Miscellaneous:
RL1 - 12V SPDT relay
CON1, CON2 - 2-pin connectors
BATT.1 - 12V battery/12V SMPS supply
LOAD - Anything like geyser or water pump

When the input threshold of IC3A is reached, its output goes low and transistor T1 cuts off. Consequently, relay RL1 de-energises, and its contact changes over to disconnect the load. Glowing LED2 indicates that the relay is energised, while D2 protects T1 from the relay coil's back-EMF when the relay is switched off.

Capacitor C2 prevents relay chatter while diode D1 resets the circuit when power is switched off by discharging the capacitor C1 through R2 and R3. LED1 indicates that the 12V supply is present. The timing is mainly determined by R2, VR1, and C1. Therefore, adjusting VR1 changes the delay. The relay's SPDT contacts (CON1/CON2) provide electrical isolation between the low-voltage timer circuit and the 230V AC load. EFY

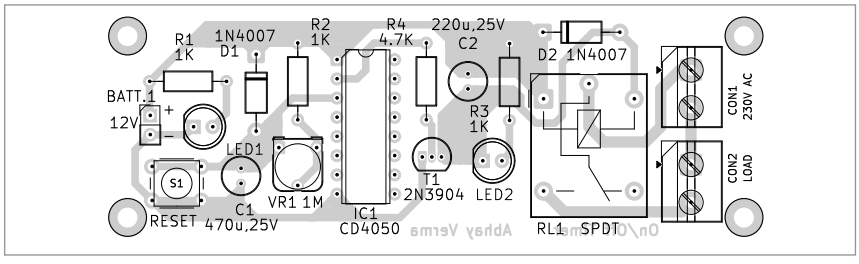


Fig. 6: Components layout

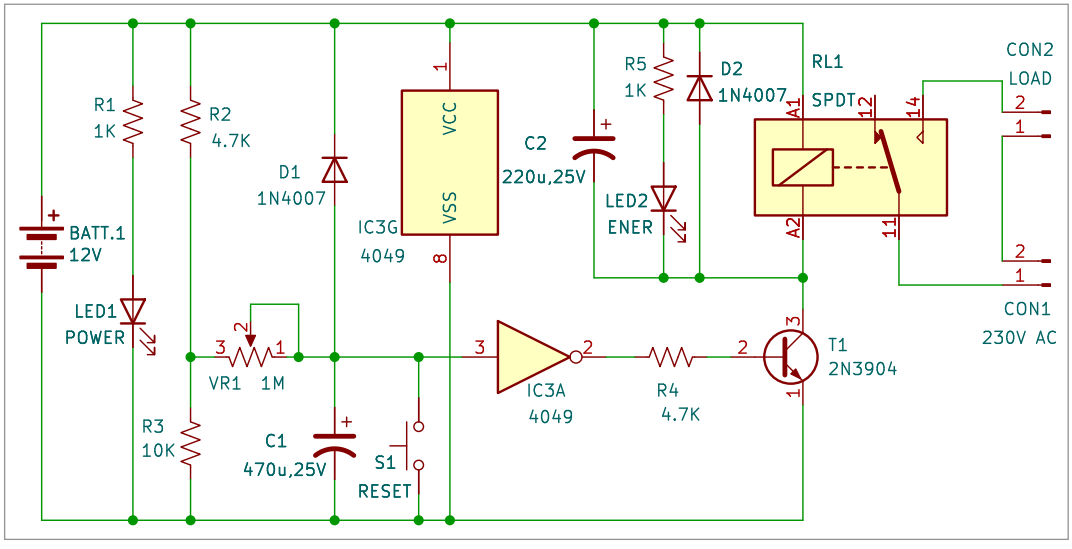


Fig. 7: Alternative modified off-delay timer

Pradeep Vasudeva is a member of the Indian Forest Service, currently posted as Director State Forest Research Institute, Jabalpur. Electronics has been his passion since adolescence, and his areas of interest include amateur radio, RF circuits, and audio projects.

